MONASH University

Expert advice from experts

Professor Marie Curie
Nobel Prize, PhD

Dr Pierre Curie
Nobel Prize, PhD

Report for
Acme Corporation

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MONASH
BUSINESS
SCHOOL

Department of Econometrics & Business Statistics

(03) 9905 2478
BusEco-Econometrics@monash.edu

ABN: 12 377 614 012

AACSB ACCREDITED EQUIS ACCREDITED ASSOCIATION AMBA ACCREDITED

1 Introduction

Kernel density estimation is a non-parametric way to estimate the probability density function of a random variable (Rice, 2007, p390). The kernel density estimate (KDE) is defined as:

$$\hat{f}(x) = \frac{1}{nh} \sum_{i=1}^{n} K\left(\frac{x - x_i}{h}\right), \tag{1}$$

where K is the kernel function, h is the bandwidth, and $x_1, x_2, \ldots, x_n$ are the observed data points. A common kernel function is the Gaussian kernel, defined as:

$$K(u) = \frac{1}{\sqrt{2\pi}} e^{-u^2/2}.$$

Figure 1 shows a KDE computed using Equation 1 with a Gaussian kernel, and h chosen using the Sheather-Jones method (Sheather & Jones, 1991).

```
library(tidyverse)
set.seed(2022 - 12 - 20)
df ← tibble(x = rnorm(200))
df ▷
  ggplot(aes(x = x)) +
  geom_density(bw = "sj") +
  geom_rug()
```

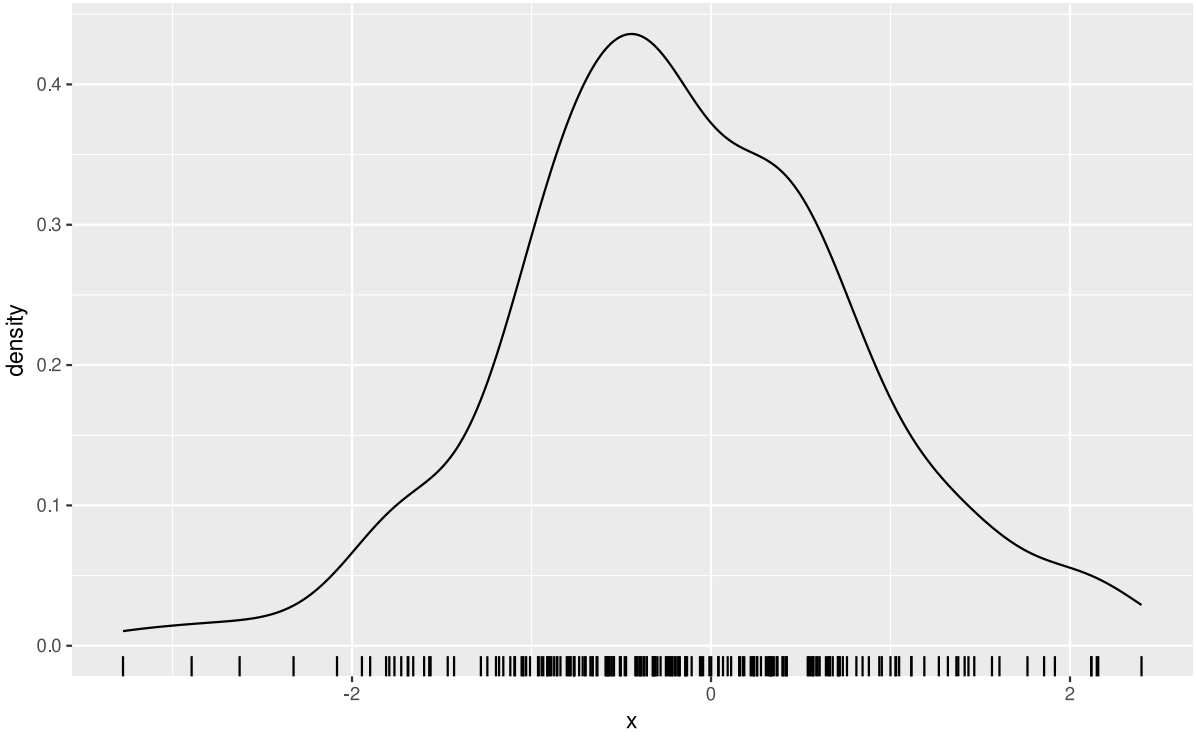


Figure 1: *A kernel density estimate obtained using 200 simulated data points from a N(0,1) distribution.*

References

Rice, J. A. (2007). *Mathematical Statistics and Data Analysis* (3rd edition). Duxbury.

Sheather, S. J., & Jones, M. C. (1991). A reliable data-based bandwidth selection method for kernel density estimation. *Journal of the Royal Statistical Society, Series B*, *53*(3), 683–690. https://doi.org/10.1111/j.2517-6161.1991.tb01857.x